

RELEASE NOTES FOR JULY 2026 UPDATE

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In this release, we extend the factors and testing portfolios through December 2025, so that the main sample period runs from January 1967 to December 2025. We also expand the testing portfolios from 199 to 201 anomalies, grouped into six categories: momentum (43), value-versus-growth (32), investment (32), profitability (50), intangibles (33), and frictions (11).

Anomaly Selection

The anomalies are a subset of the 452 anomalies in Hou, Xue, and Zhang (2020).² We first include the 158 anomalies that are significant ($|t| \geq 1.96$) in their original sample from January 1967 to December 2016. We then add anomalies that have become significant in subsequent sample extensions, including 11 anomalies for the period through December 2018 (Hou, Mo, Xue, and Zhang 2021), one through December 2019 (Rca), two through December 2020 (Sim12 and Tbi^q6), one through December 2021 (Roa6), five through December 2022 (Cm6, Ia^q1, dLti, Nxf, and Dls1), two through December 2023 (Rna^q12 and Pmq1), four through December 2024 (Gla, Ole^q12, Vcfl, and Ala^q1), and two through December 2025 (Tes1 and Tail12).³ We also include expected growth from the expected growth factor of Hou, Mo, Xue, and Zhang (2021), with 1-, 6-, and 12-month holding periods.

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² Hou, Kewei, Chen Xue, and Lu Zhang, 2020, Replicating anomalies, *Review of Financial Studies* 33, 2019-2133.

³ Hou, Kewei, Haitao Mo, Chen Xue, and Lu Zhang, 2021, An augmented q-factor model with expected growth, *Review of Finance* 25, 1-41. The 11 anomalies from the December 2018 extension are prior 11-month returns with a 12-month holding period ($R^{11}12$), 52-week high with a 12-month holding period (52w12), segment momentum with a 12-month holding period (Sm12), asset turnover (Ato), capital turnover (Cto), quarterly O-score with a 1-month holding period (O^q1), quarterly taxable income-to-book income with a 12-month holding period (Tbi^q12), quarterly sales growth with a 1-month holding period (Sg^q1), quarterly R&D expense-to-sales with 6- and 12-month holding periods (Rds^q6 and Rds^q12), as well as the nonannual component of year 1-lagged return (R_n^1).

We include 12 anomalies that, albeit insignificant, are prominent in the empirical asset pricing literature, including standardized unexpected earnings (6-month holding period, Sue6), long-term reversal (1-month holding period, Rev1), dividend yield (Dp), payout yield (Op), total accruals (Ta), operating profits-to-equity (Ope), market equity (Me), idiosyncratic volatility per the Fama-French 3-factor model (1-month holding period, Ivff1), idiosyncratic volatility per the Hou-Xue-Zhang q -factor model (1-month holding period, Ivq1), total volatility (1-month holding period, Tv1), market beta (1-month holding period, $\beta 1$), and short-term reversal (Srev).

Finally, for anomalies that have become insignificant over time since the first release of our data library, we continue their coverage to maintain backward compatibility with prior releases.

The Number of Significant Anomalies

As Table A shows, among the 452 anomalies in Hou, Xue, and Zhang (2020), the number significant at the 5% level falls from 158 in 1967-2016 to 145 in 1967-2025. The decline is concentrated in two categories: value-versus-growth, where the count drops from 29 to 9, and investment, where it falls from 28 to 22. Profitability moves in the opposite direction, rising from 35 to 44. The remaining categories are more stable.

Technical Documents

We have updated the Technical Documents that detail the construction of our factors and testing portfolios. Three changes are notable. First, we now use the new CRSP CIZ data, which constructs monthly stock returns differently and introduces many new flag variables containing stock background and event information.⁴ We accordingly adopt new variables for sample construction and revise the anomaly variables built from CRSP data.⁵ Second, we have revised our delisting

⁴ The Wharton Research Data Services (WRDS) provides a detailed [overview of the CRSP data changes](#).

⁵ We have made adjustments to the following anomaly variables: prior six-month returns (R^6), prior 11-month returns (R^{11}), six-month residual momentum (ϵ^6), 11-month residual momentum (ϵ^{11}), reversal (Rev), dividend yield (Dp), intangible return (Ir), composite equity issuance (Cei), failure probability (Fp), seasonality ($R_a^1, R_n^1, R_a^{[2,5]}, R_n^{[2,5]}, R_a^{[6,10]}, R_n^{[6,10]}, R_a^{[11,15]}, R_n^{[11,15]}, R_a^{[16,20]}, R_n^{[16,20]}$), market beta (β), tail risk (Tail), and short-term reversal (Srev). Most of these adjustments impose monthly return flag restrictions to ensure that return-based variables are measured accurately: in the CIZ data, a non-missing monthly return may span a window other than month-end to month-end, or may be compounded across unbridged gaps in daily prices. For industry- and portfolio-based anomaly variables such as industry momentum, these adjustments operate only indirectly and have minimal effect, so we leave their construction unchanged.

adjustment, both to address the incomplete delisting adjustment embedded in the new CRSP monthly returns and to accommodate other changes in the CRSP data. Third, we have added descriptions of the testing portfolios for two anomalies, Tes1 and Tail12, that have become significant in the latest sample period.

Table A: Number of Significant Anomalies							
Period	Mom (57)	V-G (69)	Inv (38)	Prof (79)	Intan (103)	Fric (106)	Total (452)
1967-2016	36	29	28	35	26	4	158
1967-2018	39	15	26	40	27	3	150
1967-2019	39	13	24	38	26	2	142
1967-2020	39	5	26	35	30	3	138
1967-2021	37	10	25	39	28	2	141
1967-2022	39	15	30	41	26	3	154
1967-2023	37	12	25	40	26	2	142
1967-2024	39	9	21	44	26	3	142
1967-2025	39	9	22	44	28	3	145

Next Update

We plan to update our data in March 2027, with the sample extended through December 2026.